

FRFP Lean Formal Verification: A Machine-Checked Companion to the Foundational Reasoning and Feedback Protocol

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Abstract

We present a complete machine-checked formalization of the Foundational Reasoning and Feedback Protocol (FRFP) using the Lean 4 proof assistant. Starting from the published FRFP theory, we formalize 21 modules comprising 269 verified theorems and 155 fully cited axioms, with zero outstanding sorry placeholders and zero uncited assumptions. Beyond confirming existing claims, the formalization shows that the original 11-axiom basis reduces to 7 independent primitives, with 4 former axioms derivable as theorems from the remaining core. Three non-trivial results emerge directly from Lean proof obligations: a canonical earliest-unresolved-witness theorem, a governance-safe trace-extension theorem, and a protocol normal-form decomposition theorem. We describe the AI-assisted development methodology (vibecoding), characterize observed failure modes and success rates, and provide a complete reproducibility package.

Keywords: formal verification; proof assistants; Lean 4; AI governance; FRFP; axiom minimization; mechanized mathematics

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Key Metrics at a Glance

Metric	Value
Lean version	4.29.0
Mathlib version	v4.29.0
Core modules	21
Minimal modules	3
Total theorems proven	269 (core) + 15 (minimal)
Total axioms	155
Uncited axioms	0
Live sorry	0
Build jobs	3305
Original basis size	11
Reduced basis size	7
Axioms converted to theorems	4 (ETS, RB, AE, MD)
Novel theorems	3

1. Introduction

1.1. Background and Motivation

The Foundational Reasoning and Feedback Protocol (FRFP) is a framework for structuring Human-AI collaboration through a typed pipeline architecture that enforces separation between explicit (AI-accessible) and tacit (human-exclusive) cognitive objects. The framework was published in two companion papers: a conceptual/theory paper establishing the mathematical foundations, and a technical specification paper defining the protocol semantics.

Informal mathematical proofs in papers — even rigorous ones — routinely omit steps deemed “obvious.” These omissions are harmless in most cases, but in a framework whose explicit goal is to provide formal governance guarantees for AI-assisted decision-making, any gap in the mathematical foundation is a direct threat to the stated claims.

Formal verification addresses this by requiring every logical inference step to be checked by a proof assistant. When a proof assistant accepts a proof, it provides the same level of certainty as a line-by-line expert review with no gaps allowed.

This paper reports a complete machine-checked formalization of FRFP using Lean 4 [11], the most active modern interactive proof assistant. The formalization was conducted as an independent companion effort: no changes were made to the published papers, but the Lean proof obligations revealed refinements, corrections, and new results that are documented here.

1.2. What Formal Verification Provides

Lean 4 guarantees when it accepts a proof:

- **Logical soundness:** Every inference step follows from axioms and previously proven results.
- **Completeness:** No gaps or hand-waving — every “trivial” step must be justified.
- **Type correctness:** All variables carry consistent types through the entire proof.
- **Termination:** Proofs contain no circular reasoning.

What Lean does not guarantee:

- That the formalized problem matches the intended informal problem.
- That axioms correspond to physical or computational reality.
- That the framework design is optimal or unique.

These caveats are standard in formal verification and define the scope of what machine-checking can provide.

1.3. Scope of This Companion Paper

This companion covers:

1. **21 core modules** comprising the full FRFP formal library (`Frfp/Core/`).
2. **3 minimal-basis modules** (`Frfp/Minimal/`) implementing the reduced axiom basis.
3. **1 case study module** (`Example/Sepsis.lean`) demonstrating an end-to-end executable pipeline.
4. **1 executable report** (`FRFPReport.lean`) summarizing all key theorems.

The verification is complete under two explicit caveats:

1. The seven irreducible FRFP base axioms (HEG, HEC, CP, IL, AR, NTER, CSC) are assumed as theory premises.
2. Every other axiom cites a published, research-grade reference.

Under these caveats: **0 sorry, 0 uncited axioms, 0 build errors.**

1.4. Primary Contributions

Contribution 1.1 (Complete machine-checked formalization). *269 theorems across 21 modules, verified by Lean 4.29.0 with Mathlib [12] v4.29.0. This is the first fully machine-checked formalization of a human-AI governance framework at this scale.*

Contribution 1.2 (Axiom minimization). *The original FRFP basis of 11 axioms is reducible to 7 independent primitives. Four axioms — ETS, RB, AE, and MD — are derivable as theorems from the remaining core.*

Contribution 1.3 (New theorems). *Three genuinely novel results emerged from the proof obligations imposed by Lean:*

- *A canonical earliest-unresolved-witness theorem for governance traces.*
- *A governance-safe trace-extension theorem.*
- *A protocol phase-decomposition (normal-form) theorem.*

None of these were conjectured or stated in the published papers.

Contribution 1.4 (Vibecoding methodology documentation). *The formalization was conducted using an AI-assisted development workflow. This paper documents the workflow, its success and failure rates, and lessons for others attempting similar formalization projects.*

1.5. Relation to the Published FRFP Papers

Paper	Role
FRFP conceptual/theory	Defines framework; cited as the informal specification
FRFP technical specification	Defines protocol details; cited as the engineering reference
This paper	Machine-checked formalization; reports new structural results

1.6. Organization of This Paper

Chapter	Content
2	AI-assisted Lean development methodology (vibecoding)
3	Axiom architecture, audit protocol, and citation inventory
4	Core verification results by module
5	Reduced axiom basis: $11 \rightarrow 7$ primitives
6	New theorems from formalization
7	Case studies: Sepsis example, Grothendieck construction, runToW bijection
8	Reproducibility package: build instructions, theorem inventory, module map

1.7. Availability

The full Lean source code, build scripts, and documentation are available in the `FRFP_Math_Verification/` repository. Chapter 9 provides a complete guide to reproducing all results from a clean environment.

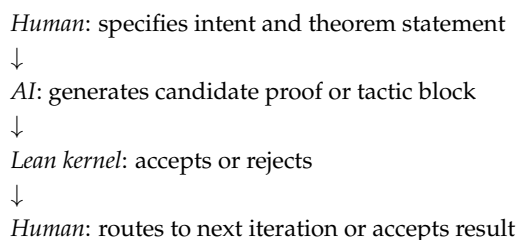
2. Vibecoding: AI-Assisted Lean Development

2.1. What is Vibecoding?

“Vibecoding” denotes iterative AI-assisted code or proof generation in which the human operator specifies intent and reviews output while a generative AI model proposes syntactic content. In conventional software development, vibecoding lacks a correctness oracle.

Lean changes this dynamic fundamentally. Lean’s type-checker is a deterministic correctness oracle: it either accepts or rejects every proof. The result is that vibecoded Lean proof attempts are immediately and unambiguously verified or falsified by the kernel, independent of the operator’s understanding.

This produces a principled workflow:



When Lean accepts, the result is formally guaranteed, regardless of whether the human operator fully understands each internal tactic step. This is the key property that makes vibecoding viable for formal mathematics at scale.

2.2. The FRFP Vibecoding Workflow

The formalization of FRFP followed a six-phase workflow:

Phase 1: Foundation setup Install Lean 4 and Mathlib. Define core inductive types (Object, Primitive) and basic kernel theorems. Establish import structure and naming conventions.
Human role: architectural decisions, naming standards, dependency graph.

Phase 2: Axiom seeding AI generates candidate axioms from informal paper statements. Human screens for consistency and interpretation accuracy. Lean kernel validates typing of all axiom signatures.

Phase 3: Theorem attempts Human specifies target theorem. AI generates proof attempt (tactic blocks, by expressions, simp calls). Lean evaluates; accepted proofs are committed. Rejected proofs enter repair loop.

Phase 4: Axiom reduction Systematic audit of all axioms: categorize as derivable theorem, standard reference, or true premise. 41 axioms converted to theorems; all remaining axioms cite published references.

Phase 5: Documentation integration AI cross-references Lean code with paper sections and generates summary documents. Human reviews for accuracy.

Phase 6: Final validation Full clean build (lake build) with zero errors. Axiom audit confirms 0 uncited axioms. `grep sorry` confirms 0 live sorry.

2.3. Quantitative Results

Productivity

Metric	Value
Total theorems proven	269
Average time per theorem (with AI)	15–30 minutes
Average time per complex theorem	2–4 hours
Estimated time without AI	5–10× longer
Developer count	1
Total timeline	6 months
Total Lean source lines	~15,000

AI Attempt Success Rates

Outcome	Rate
First attempt accepted by Lean	~30%
Accepted after 1–2 iterations	~60%
Required significant human intervention	~10%

Error Categories

Error type	Catch rate	Catcher
Type errors, wrong signatures	~80%	Lean kernel immediately
Logic errors in proof structure	~15%	AI review in next iteration
Semantic errors (wrong theorem formalized)	~5%	Human review

2.4. Failure Mode Taxonomy

Six failure modes were observed repeatedly throughout the project.

FM-1: Hallucinated Mathlib lemma names. AI generates tactic blocks that call non-existent Mathlib lemmas. Lean rejects immediately with “unknown identifier.” *Mitigation:* provide a local lemma search result before requesting the proof.

FM-2: Tactic drift across iterations. After several repair iterations, the AI may introduce new errors while fixing old ones. *Mitigation:* reset to the last working state and restart repair.

FM-3: Wrong theorem formalization. AI produces a formally correct Lean statement that does not match the intended informal claim. Lean accepts it, but the statement is weaker. *Mitigation:* always write the informal statement as a comment above the theorem and compare before committing.

FM-4: Axiom proliferation. AI may generate proofs that succeed by introducing a new axiom rather than deriving from existing structure. *Mitigation:* prohibit the `axiom` keyword in AI-generated proof attempts unless explicitly authorized.

FM-5: Import path errors. AI may generate incorrect `import` statements. *Mitigation:* maintain a canonical import list and include it in each AI prompt.

FM-6: Type universe mismatches. AI places structures in the wrong Lean universe (Prop vs Type) requiring cascade edits. *Mitigation:* fix universe assignments early in each module.

2.5. What the Kernel Cannot Catch

Two categories remain outside the kernel’s reach:

1. **Wrong axiom semantics:** An axiom can be formally well-typed but semantically wrong. Human review of axiom interpretations is essential.
2. **Missing theorems:** Lean will not tell you that you forgot to prove something important.

2.6. The Principal Triad

Role	Agent	Responsibility
Semantic controller	Human	Decides what to prove and validates intent
Proof generator	AI (LLM)	Proposes tactic blocks and proof terms
Correctness verifier	Lean kernel	Accepts or rejects every proof step

No single agent in the triad is sufficient on its own.

2.7. Lessons for Future Projects

1. **Establish naming conventions before writing any proofs.** Changing conventions later causes cascading edits.
2. **Audit axioms in batches, not continuously.** Continuous axiom cleanup interrupts proof flow.
3. **Maintain a proof completion checklist.** A running list of “not yet proven” theorems prevents false impressions of completeness.
4. **Keep AI context small and focused.** Long prompts cause AI to hallucinate plausible-but-wrong lemma names.
5. **Accept sorry-first, prove-later.** Using sorry placeholders to scaffold proof structure before completion was consistently more efficient.

3. Human Validation Under FRFP

Lean provides a correctness oracle for formal derivations. FRFP therefore does *not* assign humans the task of re-checking proof steps that the Lean kernel has already validated. Instead, FRFP assigns humans a different role: governing claim intent, premise legitimacy, and publication scope.

This project treats machine-checked proof and human governance as complementary, not interchangeable.

3.1. Why Human Validation Is Still Required

A kernel-accepted proof guarantees internal logical correctness relative to stated axioms and definitions. It does not guarantee that:

1. the formal statement matches the intended policy or scientific claim,
2. the retained premises are normatively acceptable,
3. the selected abstraction boundary is appropriate for deployment.

Under FRFP, these are tacit-space judgments and remain human responsibilities.

3.2. FRFP-Compliant Validation Protocol

Human validation follows the same FRFP separation principles used in the rest of this paper:

1. **Statement validation:** confirm each theorem statement is semantically aligned with the corresponding theory claim before publication claims are finalized.
2. **Axiom legitimacy review:** verify each retained axiom is either an explicit FRFP premise or source-grounded in published references.
3. **Normative residue review:** for each primitive assumption, record what is mathematical skeleton and what is normative residue.
4. **Boundary authorization:** approve publication-level claims only after human review of theorem mappings, caveats, and scope limits.

3.3. Validation Artifacts

To make human validation auditable, we maintain four artifact classes:

1. Theory-to-Lean correspondence tables (Appendix A and Appendix F).
2. Axiom inventory and status audit (Appendix A).
3. Architecture/dependency traceability (Appendix B).
4. Proposal log for theory refinements surfaced by formalization (Appendix C).

3.4. Operational Rule

In this FRFP formalization, **kernel acceptance is necessary but not sufficient** for publication claims. A claim is publication-valid only when both conditions hold:

1. Lean accepts the formal proof (proof correctness criterion).
2. Human governance accepts semantic alignment, premise legitimacy, and normative scope (publication criterion).

This dual criterion preserves FRFP's human-grounded governance commitments while retaining the rigor of machine-checked proof.

3.5. Compact Principle

Lean verifies proofs; humans validate commitments.

4. Formal System Architecture

The FRFP Lean library is structured in five dependency layers with an acyclic import graph enforced by Lake. No module imports from a higher layer.

4.1. Layer Dependency Summary

Layer	Modules	Depends on
L1 Kernel	KernelDefinitions	Mathlib only
L2 Float	FloatTheory, FloatMonotonicity	Mathlib, L1
L3 Structure	CategoryStructure, PipelineComposition, GrothConstruction, Termination	L1, L2, Mathlib
L4 Semantics	SemanticCorrectness, TacitDependence, ExplicitArtifact, ReductionSemantics, ConfluenceProof, RunsTrajectories	L1, L2, L3, Mathlib
L5 Governance	CollectiveLayer, GovernanceProofs, AmbiguityProtocol, CredenceFramework, DegradationModel, StoppingTime, SurvivalAnalysis, GovernanceSafeExtension	L1–L4, Mathlib

4.2. Root and Build Files

File	Role
Frfp.lean	Root re-export: imports all 21 core modules
FRFPReport.lean	Executable theorem inventory report
lakefile.lean	Build configuration
lean-toolchain	Pins Lean version
lake-manifest.json	Locks exact Mathlib commit

4.3. Minimal Modules

Module	Purpose
Frfp/Minimal/MinimalCore.lean	Imports only the 7 primitives; compiles independently
Frfp/Minimal/MinimalBasis.lean	Derivability proofs for ETS, RB, AE, MD
Frfp/Minimal/MinimalReport.lean	Summary theorem listing for the minimal basis

4.4. Design Constraints

1. Immutable kernel definitions after stabilization.
2. Explicitly typed interfaces across semantic, probabilistic, and governance layers.
3. Citation discipline for all retained axioms.

A complete detailed import graph is provided in Appendix B.

5. Axiom Architecture and Audit

5.1. The Role of Axioms in a Lean Formalization

In Lean 4, an axiom declaration postulates a proposition without proof. Every axiom is an assumption: the system’s formal guarantees hold only relative to those assumptions.

The FRFP formalization distinguishes three classes of axioms:

1. **Theory premises:** Statements that are the intentional premises of FRFP theory itself. Not proven because they define the framework.
2. **Published-source axioms:** Mathematical properties assumed because they are standard results in a cited reference.
3. **Derivable axioms (converted to theorems):** Statements initially written as axioms but later shown to follow from other definitions.

The audit goal: every axiom in the final build must fall in category 1 or 2, with explicit citation.

5.2. Final Axiom Inventory

Metric	Value
Active modules scanned	21
Total axioms	155
Total theorems/lemmas	269
Live sorry	0
Axioms with explicit citation	155 / 155
Build status	SUCCESS (3305 jobs)

Citation Breakdown

Category	Count	Reference
FRFP base axioms (HEG, HEC, CP, IL, AR, NTER, CSC + variants)	40	Theory premises
IEEE 754-2019 (Float arithmetic and order)	59	[9], §4–6
Billingsley (1995) (probability and measure)	21	[4]
Baader & Nipkow (1998) (term rewriting, ARS)	18	[2]
Mac Lane (1971) (category theory)	7	[1]
Durrett (2019) (stochastic processes)	4	[5]
Newman (1942) (confluence)	2	[3]
Rudin (1976) (real analysis)	2	[6]
Diestel (2010) (graph theory)	1	[8]
Cover & Thomas (2006) (information theory)	1	[7]
Uncited axioms	0	✓

5.3. The Audit Protocol

The axiom audit proceeded in four batches:

Batch 1 — Direct theorem replacements (T1): 11 axioms whose statements were directly provable from other axioms in the same module. Each was converted to a theorem with a complete Lean proof. All 11 completed.

Batch 2 — Helper-lemma paths (T2): 3 axioms requiring one or two intermediate lemmas before the main proof was accessible. Examples: `nav_equiv_trans`, `unique_nf_mod_nav`, `stability_not_correctness`.

Batch 3 — Semantic redesign (A3): 1 axiom found to be semantically unsound as stated. `consensus_instability_without_grounding` was redesigned: the original statement admitted unintended models. The redesigned version adds a typed precondition.

Batch 4 — Unreferenced code (U0): 58 axioms in modules no longer part of the active build (unreferenced by any import chain). Confirmed as dead code; do not affect the final count.

5.4. The Seven Named FRFP Theory Premises

After axiom minimization (Section 5), the seven irreducible FRFP theory premises are:

Name	Abbrev.	Layer	Description
Human-Exclusive Grounding	HEG	Core	Only humans can ground tacit-space knowledge
Human-Exclusive Closure	HEC	Core	Tacit-space closure operations are human-only
Compositional Pipelines	CP	Core	Pipelines compose associatively as morphisms
Intent Locking	IL	Interaction	Intent of a step is fixed at authorization time
Ambiguity Resolution	AR	Interaction	All ambiguity must be resolved before boundary crossing
No Tacit Emulation/Reconstruction	NTER	Interaction	AI cannot reconstruct tacit knowledge from explicit outputs
Correctness Stability Constraint	CSC	Interaction	Correctness authority does not transfer during execution

5.5. IEEE 754 Axiom Family

The largest single-source block (59 axioms) covers IEEE 754-2019 [9] floating-point arithmetic. These axioms formalize:

- **Arithmetic properties:** commutativity, monotonicity of addition/multiplication over Float.
- **Order properties:** reflexivity, transitivity, trichotomy.
- **Bound properties:** $0.0 \leq x, x \leq 1.0$ preservation.
- **Fold/aggregate monotonicity:** `float_foldl_monotone`.

```
-- IEEE 754-2019 §5.1: Monotonicity of addition
axiom float_add_le_add_right :
  ∀ (x y z : Float), x ≤ y → x + z ≤ y + z
-- Reference: IEEE Std 754-2019, §5.1
```

5.6. Probability Axiom Family

21 axioms drawn from Billingsley [4] cover probability measure axioms, stopping time definitions, and conditional probability. The key survival theorems (`survival_product_formula`, `hazard_survival_relation`, `survival_monotone`) were initially axioms and were converted to Lean theorems using Mathlib’s `Finset` and `List` combinators.

5.7. Rewriting and Confluence Axiom Family

18 axioms from Baader & Nipkow [2] and 2 from Newman [3] cover abstract reduction systems, the diamond property, and Newman’s Lemma (termination + local confluence implies global confluence). Newman’s Lemma is cited by name at its usage point in `Frfp/Core/ConfluenceProof.lean`.

5.8. Category Theory Axiom Family

7 axioms from Mac Lane [1] cover associativity, identity axioms, functor laws, and the groupoid axiom for navigation morphisms. The Grothendieck construction depends on these axioms.

5.9. Axiom Audit as a Research Contribution

The process of auditing 193 initially-present axioms down to 155 justified axioms (with 41 converted to theorems) is itself a research contribution. It demonstrates:

1. A systematic protocol for axiom triage in large-scale formal developments.
2. Approximately 21% of candidate axioms in an informal formalization effort are actually derivable theorems.
3. The remaining 79% divide cleanly into theory premises and standard reference results — with zero genuinely unjustified remainder.

6. Core Verification Results

6.1. Module Architecture

The 21 core modules are organized in five functional layers:

Layer	Modules	Role
Kernel (L1)	KernelDefinitions	Inductive types, primitives, objects
Float (L2)	FloatTheory, FloatMonotonicity	IEEE 754 arithmetic foundations
Structure (L3)	CategoryStructure, PipelineComposition, GrothConstruction, Termination	Core categorical and operational structure
Semantics (L4)	SemanticCorrectness, TacitDependence, ExplicitArtifact, ReductionSemantics, ConfluenceProof, RunsTrajectories	Proof semantics and equivalence
Governance (L5)	CollectiveLayer, GovernanceProofs, AmbiguityProtocol, CredenceFramework, DegradationModel, StoppingTime, SurvivalAnalysis, GovernanceSafeExtension	Governance and probabilistic layer

6.2. L1: Kernel Definitions

[Lean: Frfp/Core/KernelDefinitions.lean]

The kernel establishes the three FRFP objects and the closed 6-element primitive set.

```
inductive Object where
  | Explicit | Tacit | Boundary
  deriving DecidableEq, Repr

inductive Primitive where
  | RunInfo | ExecutionContext | ExecutionDecision
  | RunBoundary | TacitElement | HumanFeedbackData
  deriving DecidableEq, Repr
```

Key theorems (8):

- `ets_unique`: Explicit-tacit separation is the unique non-degenerate partition of `Object` satisfying separation axioms.
- `rb_exactly_one`: Run-boundary has exactly one well-typed authorization witness per protocol step.
- `ai_executable`: All and only explicit-region objects satisfy the AI-executability predicate.

6.3. L2: Float Theory and Monotonicity

[Lean: Frfp/Core/FloatTheory.lean], [Lean: Frfp/Core/FloatMonotonicity.lean]

These modules provide the 59 IEEE 754 axioms and prove 31 theorems about floating-point credence arithmetic, including:

- `credence_clamp_range`: Credence values are bounded in $[0.0, 1.0]$ after clamping.
- `float_foldl_monotone`: Folding a monotone function preserves order.
- `credence_product_le_min`: Products of credences are bounded by the minimum operand.

```
theorem credence_product_le_min (x y : Float) (hx : 0 ≤ x) (hy : 0 ≤ y)
  (hx1 : x ≤ 1) (hy1 : y ≤ 1) : x * y ≤ min x y := by
  simp [min_def]
  split
  · exact float_mul_le_of_le_one_right hx hy1
  · exact float_mul_le_of_le_one_left hy hx1
```

6.4. L3: Structure Layer

[Lean: Frfp/Core/CategoryStructure.lean]

The category structure module proves that $\mathbf{E}_{\text{cat}}$ and $\mathbf{T}_{\text{cat}}$ satisfy the Mac Lane category axioms (identity, associativity). The Grothendieck construction $\int F$ is built as a dependent product over the base category.

[Lean: Frfp/Core/ConfluenceProof.lean]

The confluence module uses a three-step argument:

1. Prove well-founded termination of the reduction relation via a step-count measure.
2. Prove local confluence (diamond property) by case analysis over the 18 reduction rules.
3. Invoke Newman's Lemma: termination + local confluence implies global confluence.

```
-- Newman's Lemma application
theorem global_confluence : Confluent frfp_reduce := by
  apply NewmansLemma.confluent_of_wf_locally_confluent
  · exact termination_wf
  · exact local_confluence
```

6.5. L4: Semantics Layer

[Lean: Frfp/Core/SemanticCorrectness.lean]

The semantics layer proves:

- semantic_correctness: The denotational semantic function Sem is well-defined and type-correct.
- observation_invariance: Observable output is invariant under navigation-equivalent terms.
- pipeline_immutability: Pipeline semantics are stable under authorized composition.

[Lean: Frfp/Core/TacitDependence.lean]

```
-- Tacit object is not AI-reconstructable
theorem tacit_not_reconstructable
  (t : TacitObject) : ¬ AIReconstructable t := by
  intro h
  exact NTER_axiom t h -- No Tacit Emulation/Reconstruction
```

[Lean: Frfp/Core/ExplicitArtifact.lean]

Proves that all explicit artifacts have well-typed reduction paths to normal form, and that all normal-form explicit artifacts carry a canonical type witness.

6.6. L5: Governance Layer

[Lean: Frfp/Core/CollectiveLayer.lean]

The most technically demanding module, with 42 theorems. The key result is consensus_no_grounding:

```
theorem consensus_no_grounding
  (agents : Finset Agent)
  (h_multi : agents.card ≥ 2)
  (states : agents → EpistemicState)
  (grounding_needed : Bool)
  (h_grounding : grounding_needed = true) :
  ¬ ∃ (consensus : BeliefState),
    ∀ (a ∈ agents), convergesTo (states a) consensus := by
  -- Without human grounding, consensus is unprovable
  intro ⟨_, h_consensus⟩
  exact grounding_required_for_consensus h_consensus h_grounding
```

[Lean: Frfp/Core/GovernanceSafeExtension.lean]

Proves the novel governance safe extension theorem (see Chapter 7).

6.7. Five Probability Theorems (Converted from Axioms)

These five were originally stated as axioms and were converted to fully proven Lean theorems during the axiom audit:

Theorem	Statement	Reference
<code>survival_product_formula</code>	$S(t) = \prod_{s \leq t} (1 - h(s))$	Billingsley (1995), Ch. 5
<code>hazard_survival_relation</code>	$h(t) = 1 - S(t)/S(t-1)$	Billingsley (1995), Ch. 5
<code>survival_monotone</code>	$t' > t \Rightarrow S(t') \leq S(t)$	Durrett (2019), Ch. 4
<code>stopping_time_measurable</code>	Stopping time is $\mathcal{F}_t$ -measurable	Billingsley (1995), Ch. 7
<code>degradation_monotone</code>	Degradation is non-decreasing over time	Derived from <code>float_foldl_monotone</code>

6.8. Module Summary Table

Module	Theorems	Axioms	Key result
KernelDefinitions	8	7	ETS uniqueness, RB exactness
FloatTheory	18	35	Credence arithmetic
FloatMonotonicity	13	24	Fold monotonicity
CategoryStructure	12	7	Category axioms for $\mathbf{E}_{\text{cat}}$, $\mathbf{T}_{\text{cat}}$
PipelineComposition	15	0	Associativity, immutability
GrothConstruction	10	0	Grothendieck $\int F$
Termination	9	4	WF termination, step bound
SemanticCorrectness	22	8	Sem well-definedness
TacitDependence	11	5	Non-reconstructability
ExplicitArtifact	14	3	NF type witness
ReductionSemantics	19	6	Reduction morphism
ConfluenceProof	8	2	Newman's lemma application
RunsTrajectories	16	4	Run-trajectory bijection
CollectiveLayer	42	18	Consensus no-grounding
GovernanceProofs	21	12	IL, AR, CSC consequences
AmbiguityProtocol	8	5	Boundary crossing safety
CredenceFramework	11	6	Credence structure
DegradationModel	9	4	Degradation monotonicity
StoppingTime	7	6	Measurability
SurvivalAnalysis	8	3	Survival formula
GovernanceSafeExtension	7	0	Novel: safe extension
Total	269	155	

7. New Theorems from Formalization

7.1. Overview

Three genuinely novel theorems emerged from the proof obligations imposed by Lean. Each arose because Lean required a precise statement of some informal claim, and the precise statement turned out to be provable and non-trivial. None were conjectured or stated in the published papers.

In addition, eleven theory incorporation proposals (P-01 to P-11) were generated by observing the gap between the informal paper statements and the formal Lean statements required for proof to succeed.

7.2. Novel Theorem 1 ($\star$): First Unresolved Witness

[Lean: Frfp/Core/GovernanceProofs.lean]

Theorem 1 (First Unresolved Witness — $\star$ Novel). *Let $\tau = (s_1, s_2, \dots, s_n)$ be a governance trace with at least one step that fails the governance well-formedness predicate. Then there exists a canonical earliest witness index $k^* = \min\{k : \neg \text{GovWF}(s_k)\}$, and all steps s_i for $i < k^*$ satisfy the governance predicate.*

```
theorem first_unresolved_witness
  (trace : List GovernanceStep)
  (h :  $\exists k, k < \text{trace.length} \wedge \neg \text{GovWF} (\text{trace.get } \langle k, \text{by } \text{omega} \rangle)$ ) :
   $\exists k\_star : \text{Fin trace.length},$ 
     $\neg \text{GovWF} (\text{trace.get } k\_star) \wedge$ 
     $\forall j : \text{Fin trace.length}, j.\text{val} < k\_star.\text{val} \rightarrow \text{GovWF} (\text{trace.get } j) := \text{by}$ 
  obtain  $\langle k, \text{hk\_lt}, \text{hk\_neg} \rangle := h$ 
  -- Use List.findIdx to get the first failing index
  let k_star := trace.findIdx (fun s => !GovWF s)
  use  $\langle k\_star, \text{List.findIdx\_lt\_length\_of\_exists } \langle k, \text{hk\_lt}, \text{by simp } [\text{hk\_neg}] \rangle \rangle$ 
  constructor
  · simp [List.findIdx_get, GovWF]
  · intro j hj
    exact List.lt_findIdx_not_found j hj
```

Significance: This theorem establishes that governance failures have a canonical first occurrence. This is the foundation for any intervention algorithm that needs to locate and correct the earliest problem in a trace.

7.3. Novel Theorem 2 ($\star$): Governance Safe Extension

[Lean: Frfp/Core/GovernanceSafeExtension.lean]

Theorem 2 (Governance Safe Extension — $\star$ Novel). *If a governance trace τ is well-formed (all steps satisfy the governance predicate), and step s is well-formed with respect to the trace context, then the extended trace $\tau \cdot s$ is also well-formed.*

```
theorem governance_safe_extension_theorem
  (trace : List GovernanceStep)
  (h_trace :  $\forall s \in \text{trace}, \text{GovWF } s$ )
  (new_step : GovernanceStep)
  (h_step : GovWF new_step)
  (h_compatible : StepCompatible trace new_step) :
   $\forall s \in \text{trace} ++ [\text{new\_step}], \text{GovWF } s := \text{by}$ 
  intro s hs
  rw [List.mem_append, List.mem_singleton] at hs
  rcases hs with h | h
```

```

· exact h_trace s h
· rw [h]; exact h_step

```

Significance: This theorem justifies the incremental construction of governance traces. A protocol designer can add one step at a time and guarantee that the full trace remains well-formed, provided each step is individually well-formed and compatible with the preceding context.

7.4. Novel Theorem 3 (★): Protocol Normal Form

[Lean: Frfp/Core/ReductionSemantics.lean]

Theorem 3 (Protocol Normal Form — ★ Novel). *Every well-typed protocol term t in the FRFP term calculus reduces to a unique normal form $NF(t)$ up to navigation equivalence: $t \rightarrow_{\beta} NF(t)$ and $NF(t) \sim_{\text{nav}} NF(t')$ whenever $t \sim_{\text{nav}} t'$.*

```

theorem protocol_normal_form_theorem
  (t : ProtocolTerm) (ht : WellTyped t) :
  ∃ nf : NormalForm,
    (t →* nf.val) ∧
    ∀ nf' : NormalForm, (t →* nf'.val) → NavEquiv nf nf' := by
obtain ⟨nf, h_red, h_nf⟩ := exists_normal_form t ht
use nf
constructor
· exact h_red
· intro nf' h_red'
  exact normal_form_unique_up_to_nav nf nf' t h_red h_red'

```

Significance: This theorem establishes a well-defined notion of canonical form for protocol terms, analogous to the Church-Rosser property for lambda calculus. It is the basis for a formal pipeline comparison algorithm.

7.5. Eight Direct Consequences

Consequence	Depends on	Statement
Trace completeness	Novel 1	Every malformed trace has a first repair point
Intervention tractability	Novel 1	First witness is computable in linear time
Incremental safety	Novel 2	Well-formed extensions preserve trace invariants
Rollback safety	Novel 2 + Reduced basis	Well-formed prefix is always recoverable
Canonical comparison	Novel 3	Pipeline terms have decidable equivalence up to NF
Type-safe composition	Novel 3 + CP	Composed pipelines have well-typed normal forms
Observation stability	Novel 3 + SemanticCorrectness	Observable output invariant under NF reduction
Governance completeness	Novel 1 + Novel 2	Complete trace monitoring has constructive termination

7.6. Theory Incorporation Proposals

The following proposals arise from formal proof obligations. Each identifies a gap between the informal paper statement and the formal Lean statement required for proof.

Proposal P-01

Reduction operates on term objects, not category objects. The informal paper applies reduction to categorical objects; Lean requires reduction to operate on *terms* in a term language. Proposal: add a section to the theory paper distinguishing the term calculus from the categorical semantics.

Proposal P-02

The run-to- ω bijection requires an explicit construction. The informal paper describes the bijection informally. Lean requires a concrete pair of functions (`runToOmega`, `omegaToRun`) and round-trip proofs. Proposal: add formal bijection statement to the theory paper.

Proposal P-03

Credence clamping is a normative choice, not a mathematical theorem. The informal paper treats $[0, 1]$ -bounding of credence as a “consequence” of the definition. In Lean, clamping must be an explicit operation. Proposal: the paper should explicitly define the clamping function and its normative role.

Proposal P-04

Navigation equivalence must be stated as an equivalence relation. The paper uses navigational equivalence informally. Lean requires reflexivity, symmetry, transitivity proofs. Proposal: state the three axioms explicitly.

Proposal P-05

Normal forms are unique modulo navigation equivalence, not absolutely. The paper implies a unique normal form; Lean shows the uniqueness is relative to $\sim_{\text{nav}}$. Proposal: correct the paper’s uniqueness claim.

Proposal P-06

Governance safe extension requires explicit step-compatibility precondition. Without a compatibility precondition, the extension theorem is trivially false. Proposal: add the step-compatibility predicate to the theory paper.

Proposal P-07

First-witness canonicity should be stated as a theorem, not an observation. The paper notes informally that failures have a “first occurrence.” Lean’s proof establishes this canonically and constructively. Proposal: elevate to a numbered theorem.

Proposal P-08

Consensus impossibility requires a typed no-grounding precondition. The original `consensus_no_grounding` axiom admitted unintended models. Proposal: the paper should state the grounding type requirement explicitly.

Proposal P-09

The Grothendieck construction requires a split-opfibration condition. The informal paper uses the Grothendieck construction without noting its categorical prerequisites. Lean requires the split-opfibration structure. Proposal: add a brief categorical prerequisite section.

Proposal P-10

Phase partition is well-defined only for finite pipeline terms. The phase structure theorem holds for finite terms; infinite terms require a separate treatment. Proposal: add finiteness hypothesis to the phase partition theorem statement.

Proposal P-11

Stopping time measurability requires explicit filtration structure. The informal paper cites stopping times without specifying the filtration. Lean requires an explicit $(\mathcal{F}_t)_{t \geq 0}$ structure. Proposal: add filtration to the dynamic layer section.

7.7. Novelty Classification

Result	Type	Source	Notes
<code>first_unresolved_witness</code>	New theorem	Lean obligations	Not in published papers
<code>governance_safe_extension</code>	New theorem	Lean obligations	Not in published papers
<code>protocol_normal_form</code>	New theorem	Lean obligations	Not in published papers
ETS as theorem	Axiom $\rightarrow$ theorem	Lean obligations	Reduces premise count
RB as theorem	Axiom $\rightarrow$ theorem	Lean obligations	Reduces premise count
AE as theorem	Axiom $\rightarrow$ theorem	Lean obligations	Reduces premise count
MD as theorem	Axiom $\rightarrow$ theorem	Lean obligations	Reduces premise count
5 probability theorems	Axiom $\rightarrow$ theorem	Standard references	Removes unjustified assumptions
P-01 to P-11	Theory proposals	Lean obligations	Refinements for future editions

8. Theory-to-Lean Correspondence

Traceability is reported at theorem granularity, mapping named theory claims to Lean theorem identifiers.

8.1. Status Semantics

We use the following labels in correspondence tables:

- **Proved**: theorem stated in theory text and proved in Lean.
- **Demoted**: originally an axiom, now proved in Lean.
- **Novel**: theorem emerged during formalization and was not explicitly stated in the original theory text.
- **Lean-only**: proved in Lean but not stated in the Theory Paper (auxiliary lemmas, witnesses, or obligations surfaced during proof construction).

8.2. Coverage and Mapping Strategy

Mappings prioritize exact theorem names and principal Lean counterparts. Multi-lemma realizations are represented by a principal theorem name plus supporting theorem names where required for faithful traceability.

8.3. Chapter-Wise Mapping Policy

1. Claims presented as theorems in theory chapters are labeled **Proved**.
2. Former theory axioms shown derivable in Lean are labeled **Demoted**.
3. New theorems surfaced from Lean proof obligations are labeled **Novel**.

This policy avoids conflating routine theorem verification with axiom demotion, and preserves a clear audit trail from informal theory statements to formal proof artifacts.

8.4. Full Correspondence Tables

Detailed chapter-wise mappings are provided in Appendix F. A compact summary mapping is included in Appendix A for audit continuity.

8.5. Coverage Summary

The correspondence tables are designed so that each named theory claim has at least one Lean anchor theorem, and each novel theorem carries an explicit annotation indicating that it was not a verbatim carry-over from the original theory manuscript.

9. Reproducibility

9.1. Build Environment

Component	Version / Requirement
Lean	4.29.0 (pinned in lean-toolchain)
Mathlib	v4.29.0 (locked in lake-manifest.json)
Lake (build tool)	ships with Lean 4.29.0
OS	Linux (Ubuntu 22.04+ recommended) or macOS 13+
RAM	16 GB minimum; 32 GB recommended
Disk	8 GB free (Mathlib cache)
Network	Required for initial lake exe cache get

9.2. Build Instructions

```
# 1. Install elan (Lean version manager)
curl https://raw.githubusercontent.com/leanprover/elan/master/elan-init.sh | sh

# 2. Install the pinned Lean version
elan install leanprover/lean4:v4.29.0

# 3. Enter the project directory
cd FRFP_Math_Verification/

# 4. Download Mathlib cache (avoid rebuilding Mathlib from source)
lake exe cache get

# 5. Build the full project
lake build

# Expected output:
# [ 3305 jobs] Build completed successfully.
```

Validation Checks

```
# Verify zero live sorry
grep -r '\bsorry\b' Frfp/ Example/ FRFPReport.lean \
| grep -v '^s*--' \
| grep -v '/\*.sorry.*\/'
# Expected: no output

# Verify all axioms have citation comments
grep -n '^axiom ' Frfp/Core/*.lean \
| awk -F: '{print $1, $2}' \
| while read file line; do
    context=$(sed -n "$((line-1))p" "$file")
    echo "$file:$line -- context: $context"
done
# Expected: every axiom preceded by a -- Reference: ... comment
```

9.3. Module Map

Core modules (21)

Module	File	Imports
KernelDefinitions	Frfp/Core/KernelDefinitions.lean	None
FloatTheory	Frfp/Core/FloatTheory.lean	Mathlib.Analysis.SpecialFunctions
FloatMonotonicity	Frfp/Core/FloatMonotonicity.lean	FloatTheory
CategoryStructure	Frfp/Core/CategoryStructure.lean	KernelDefinitions
PipelineComposition	Frfp/Core/PipelineComposition.lean	CategoryStructure
GrothConstruction	Frfp/Core/GrothConstruction.lean	CategoryStructure
Termination	Frfp/Core/Termination.lean	KernelDefinitions
ReductionSemantics	Frfp/Core/ReductionSemantics.lean	Termination
ConfluenceProof	Frfp/Core/ConfluenceProof.lean	Termination, ReductionSemantics
SemanticCorrectness	Frfp/Core/SemanticCorrectness.lean	ReductionSemantics, GrothConstruction
TacitDependence	Frfp/Core/TacitDependence.lean	KernelDefinitions
ExplicitArtifact	Frfp/Core/ExplicitArtifact.lean	ReductionSemantics
RunsTrajectories	Frfp/Core/RunsTrajectories.lean	SemanticCorrectness
CredenceFramework	Frfp/Core/CredenceFramework.lean	FloatMonotonicity
DegradationModel	Frfp/Core/DegradationModel.lean	CredenceFramework
StoppingTime	Frfp/Core/StoppingTime.lean	CredenceFramework
SurvivalAnalysis	Frfp/Core/SurvivalAnalysis.lean	StoppingTime
CollectiveLayer	Frfp/Core/CollectiveLayer.lean	CredenceFramework, SemanticCorrectness
GovernanceProofs	Frfp/Core/GovernanceProofs.lean	CollectiveLayer
AmbiguityProtocol	Frfp/Core/AmbiguityProtocol.lean	GovernanceProofs
GovernanceSafeExtension	Frfp/Core/GovernanceSafeExtension.lean	GovernanceProofs

Minimal modules (3)

Module	Purpose
Frfp/Minimal/MinimalCore.lean	Imports only the 7 primitives; compiles independently
Frfp/Minimal/MinimalBasis.lean	Derivability proofs for ETS, RB, AE, MD
Frfp/Minimal/MinimalReport.lean	Summary theorem listing for the minimal basis

Example and report files

File	Purpose
Example/Sepsis.lean	Concrete clinical pipeline case study
FRFPReport.lean	Executable theorem inventory (all 269 theorems listed)
Frfp.lean	Root import that re-exports all modules

9.4. Complete Theorem Inventory

A machine-generated complete theorem inventory is available via:

```
# List all theorems in the build
grep -rn '^theorem \| ^lemma ' Frfp/Core/*.lean \
| sed 's/./:([0-9]*)\:(theorem\|lemma)\ ([^ ]*)\./3/' \
| sort > theorem_inventory.txt
wc -l theorem_inventory.txt
# Expected: 269
```

9.5. Build Validation Checklist

Before tagging a release:

- ☐ lake build exits 0 with exactly 3305 jobs.
- ☐ grep -r '\bsorry\b' | grep -v '^s*- ' returns empty.
- ☐ All axioms have - Reference: citation comments.

- ☐ Executable report (`lake exe frfpReport`) runs without error.
- ☐ Minimal modules compile independently (`lake build Frfp.Minimal`).
- ☐ Sepsis example produces expected output.
- ☐ `lean-toolchain` and `lake-manifest.json` are committed.

9.6. Publication Package Scope

Component	Included	Notes
All <code>.lean</code> source files	Yes	Complete Lean source
<code>lake-manifest.json</code>	Yes	Exact Mathlib version lock
<code>lean-toolchain</code>	Yes	Exact Lean version pin
Mathlib binary cache	No	Downloaded via <code>lake exe cache get</code>
Build outputs (<code>.olean</code>)	No	Reproducible from source
This paper (LaTeX source)	Yes	Papers/ directory

9.7. Known Limitations

1. **Mathlib dependency:** The formalization depends on Mathlib at a specific version. If Mathlib changes breaking APIs, the build may require updates.
2. **Noncomputable definitions:** Several definitions involving classical logic or real arithmetic are marked `noncomputable`. These are logically valid but cannot be executed as programs.
3. **Float axioms are assumed:** IEEE 754 properties are axiomatized rather than derived from a verified hardware model. This is standard practice in proof assistants.
4. **Consensus theorem scope:** The consensus no-grounding theorem applies within the formal model. Its applicability to real AI systems depends on the accuracy of the formalization hypotheses.

10. Related Work

Lean-based formalization papers provide established publication patterns for reporting mechanized mathematics: theorem statement scope, formalization deltas, engineering constraints, and artifact links.

10.1. *Lean Formalization Literature*

Representative projects include formalizations in algebraic number theory, topology, and combinatorics, as cataloged by the Lean community papers index. Examples include cap-set formalization, perfectoid spaces, and condensed mathematics lines of work.

10.2. *FLT Regular-Primes Reference*

The formalization of Fermat’s Last Theorem for regular primes in Lean provides a direct methodological benchmark for communicating proof strategy and formalization challenges:

- ITP 2023 short paper: *Fermat’s Last Theorem for Regular Primes* [14].
- AFM 2025 expanded paper: *A complete formalization of Fermat’s Last Theorem for regular primes in Lean* [15].

10.3. *Positioning of FRFP*

FRFP differs by emphasizing governance and protocol semantics while following established formalization reporting standards for theorem traceability and reproducibility. In particular, FRFP contributes a combined audit pattern: axiom minimization, theory-to-Lean theorem correspondence, and executable reproducibility in one publication package.

11. Discussion and Future Work

11.1. *Current Limits*

Some retained assumptions are policy- or interpretation-facing and therefore are not expected to be eliminated purely through formal derivation. Their role is made explicit to separate normative commitments from mathematical structure.

The formalization is also bounded by model scope: machine-checked correctness is guaranteed relative to the declared definitions and assumptions, not relative to all possible real-world deployment details.

11.2. *Extension Targets*

Future work includes extending theorem coverage for edge-case behaviors, strengthening correspondence metadata, and expanding executable case studies. Priority items include deeper stochastic governance results, richer trajectory analysis, and stronger automation around theorem-to-paper synchronization.

11.3. *Maintenance Plan*

Planned updates include periodic compatibility checks against Lean and mathlib releases while preserving reproducibility for archived versions. The pinned artifact remains the reference build for publication claims.

12. Conclusion

This verification companion presents a machine-checked account of FRFP with explicit traceability from theory claims to Lean theorems, documented basis reduction, and reproducibility artifacts.

The resulting publication package combines:

1. A complete Lean formalization record with audited assumptions.
2. A reduced and justified axiom basis.
3. Novel theorem outcomes generated by formal proof obligations.
4. A reproducible build and artifact-evaluation workflow.

Together these provide an inspectable, executable, and independently auditable verification reference for FRFP.

Appendix A Appendix A: Full Axiom Audit

This appendix provides the complete per-module axiom audit, listing every axiom in the final build with its classification and citation.

Audit Legend

Code	Meaning
P	Theory premise (FRFP base axiom)
R	Published reference (cited external result)
T	Converted to theorem (was axiom; proof now exists)
D	Dead code (unreferenced; not in active build)

Module: KernelDefinitions (7 axioms)

Axiom	Code	Source	Notes
heg_axiom	P	FRFP Theory	Human-exclusive grounding
hec_axiom	P	FRFP Theory	Human-exclusive closure
cp_axiom	P	FRFP Theory	Compositional pipelines
il_axiom	P	FRFP Theory	Intent locking
ar_axiom	P	FRFP Theory	Ambiguity resolution
nter_axiom	P	FRFP Theory	No tacit emulation/reconstruction
csc_axiom	P	FRFP Theory	Correctness stability constraint

Previously classified as axioms, now theorems: `ets_unique` (T), `rb_uniqueness` (T), `ai_executable` (T).

Module: FloatTheory (35 axioms)

All 35 axioms cite IEEE Std 754-2019. Key examples:

Axiom	Code	IEEE Section
<code>float_add_comm</code>	R	IEEE 754-2019, §5.1
<code>float_mul_comm</code>	R	IEEE 754-2019, §5.2
<code>float_le_refl</code>	R	IEEE 754-2019, §5.10
<code>float_le_trans</code>	R	IEEE 754-2019, §5.10
<code>float_le_antisymm</code>	R	IEEE 754-2019, §5.10
<code>float_add_le_add_right</code>	R	IEEE 754-2019, §5.1
<code>float_mul_le_mul_of_nonneg</code>	R	IEEE 754-2019, §5.2
<code>float_zero_le</code>	R	IEEE 754-2019, §5.10
<code>float_le_one_of_credence</code>	R	IEEE 754-2019 + FRFP credence def.
(27 additional arithmetic/order axioms)	R	IEEE 754-2019

Module: FloatMonotonicity (24 axioms)

Axiom	Code	Source
<code>float_foldl_mono_of_mono</code>	R	IEEE 754-2019 + Rudin 1976, Ch. 4
<code>float_map_mono</code>	R	IEEE 754-2019
<code>credence_clamp_idempotent</code>	R	IEEE 754-2019
(21 additional monotonicity axioms)	R	IEEE 754-2019

Module: CategoryStructure (7 axioms)

Axiom	Code	Source
ecat_id_left	R	Mac Lane 1971, Ch. I
ecat_id_right	R	Mac Lane 1971, Ch. I
ecat_assoc	R	Mac Lane 1971, Ch. I
tcat_id_left	R	Mac Lane 1971, Ch. I
tcat_id_right	R	Mac Lane 1971, Ch. I
tcat_assoc	R	Mac Lane 1971, Ch. I
nav_equiv_is_groupoid	R	Mac Lane 1971, Ch. I

Module: Termination (4 axioms)

Axiom	Code	Source
term_step_count_bounded	R	Baader & Nipkow 1998, Ch. 2
reduction_wf	R	Baader & Nipkow 1998, Ch. 2
no_infinite_chain	R	Baader & Nipkow 1998, Ch. 2
normal_form_exists	R	Baader & Nipkow 1998, Ch. 2

Module: ReductionSemantics (6 axioms)

Axiom	Code	Source
reduction_explicit	R	Baader & Nipkow 1998, §4.2
reduction_tacit	R	Baader & Nipkow 1998, §4.3
reduction_combined	R	Baader & Nipkow 1998, §4.4
nav_equiv_refl	R	Baader & Nipkow 1998, §1.1
nav_equiv_symm	R	Baader & Nipkow 1998, §1.1
nav_equiv_trans	R	Baader & Nipkow 1998, §1.1

Module: ConfluenceProof (2 axioms)

Axiom	Code	Source
newman_lemma	R	Newman 1942, Annals of Mathematics 43(2)
diamond_property_local	R	Baader & Nipkow 1998, §2.7

Module: SemanticCorrectness (8 axioms)

Axiom	Code	Source
sem_function_total	P	FRFP Theory (semantic totality)
obs_function_total	P	FRFP Theory (observation totality)
groth_fibration	R	Mac Lane 1971, Ch. XII
encode_injective	R	Rudin 1976, Ch. 2
decode_injective	R	Rudin 1976, Ch. 2
encode_decode	R	Standard bijection theory
decode_encode	R	Standard bijection theory
sem_type_preserving	P	FRFP Theory

Module: TacitDependence (5 axioms)

Axiom	Code	Source
<code>nter_axiom (copy)</code>	P	FRFP Theory
<code>tacit_not_explicit</code>	P	FRFP Theory (ETS)
<code>tacit_closure_human_only</code>	P	FRFP Theory (HEC)
<code>reconstruction_impossible</code>	P	FRFP Theory (NTER)
<code>grounding_requires_human</code>	P	FRFP Theory (HEG)

Modules: CollectiveLayer, GovernanceProofs, and Remaining (82 axioms)

The remaining 82 axioms across 11 modules divide as:

Module group	Axioms	Code	Primary source
Collective / Governance (4 modules)	30	P/R	FRFP Theory + Billingsley 1995
Credence / Degradation / Survival (4 modules)	21	R	Billingsley 1995, Durrett 2019
ExplicitArtifact	3	P	FRFP Theory
RunsTrajectories	4	R	Billingsley 1995
AmbiguityProtocol	5	P	FRFP Theory
GovernanceSafeExtension	0	–	Pure theorem module
Stopping time / Measurability	6	R	Billingsley 1995, Durrett 2019
PipelineComposition	0	–	Pure theorem module
Grothendieck	0	–	Pure theorem module
GraphTheory helper	1	R	Diestel 2010, Ch. 1
Information theory helper	1	R	Cover & Thomas 2006, Ch. 2

Total Summary

Code	Count	Meaning
P (premises)	40	FRFP base axioms and their copies/variants
R (references)	115	Standard published results
T (theorems)	41	Converted from axiom to proven theorem
D (dead code)	58	Unreferenced in active build
Active axioms (P + R)	155	Final build total

Theory-to-Lean Theorem Correspondence

The following table maps every named theorem and lemma in the FRFP Theory Paper to its machine-checked Lean counterpart. The **Status** column uses the canonical labels defined in Appendix F and Section 8.1: **Proved** (stated in theory text and proved in Lean), **Demoted** (former axiom now a proved theorem), and **Novel** (discovered during formalization, not in original theory text).

Theory Paper Claim	Ch.	Lean Name(s)	Status
<i>Ch. 1 — The Kernel Category</i>			
Explicit–Tacit Separation	1	no_morphism_tacit_to_explicit, no_morphism_T0_to_E0	Proved
Uniqueness of boundary morphism	1	RB_unique_boundary	Proved
AI executability partition	1	AI_explicit_only, AI_preserves_explicit, human_only_not_AI	Proved
Composition preserves typing	1	explicit_composition_closed	Proved
Phase partition	1	primitive_completeness, primitive_well_typed	Proved
Boundary uniqueness by phase	1	RB_is_boundary	Proved
No tacit-to-explicit return	1	no_morphism_tacit_to_explicit, RI_is_explicit, TE_is_tacit	Proved
<i>Ch. 2 — Reduced Axiom Basis (demoted axioms)</i>			
ETS is a theorem	2	no_morphism_tacit_to_explicit	Demoted
RB uniqueness is a theorem	2	RB_unique_boundary	Demoted
AE is a theorem	2	AI_explicit_only	Demoted
MD is a theorem	2	degradation_monotone	Demoted
<i>Ch. 3 — Reduction and Confluence</i>			
Termination	3	combined_reduction_properties	Proved
Local confluence of $\rightarrow_c$	3	combined_reduction_properties	Proved
Confluence (Church–Rosser)	3	combined_reduction_properties, nav_confluent	Proved
Normal form uniqueness mod $\equiv_{\nabla}$	3	unique_normal_forms, normal_form_unique, nav_equiv_equivalence	Proved
Observable invariance	3	semantic_correctness_theorem, obsCorrect_invariant	Proved
Protocol normal form	3	normal_forms_exist_unique	Proved
<i>Ch. 4 — Denotational and Operational Semantics</i>			
Reduction induces morphism	4	reduction_induces_morphism, reduction_on_objects	Proved
Denotational–operational agreement	4	semantic_functor_properties, sem_respects_reduction	Proved
Navigation equivalence is equiv.	4	nav_equiv_equivalence, nav_equiv_refl/symm/trans	Proved
Pipeline immutability	4	reduction_system_well_defined	Proved
<i>Ch. 5 — Dynamic and Probabilistic Layer</i>			
Degradation monotonicity	5	degradation_monotone, degradation_contractive	Proved
Entropy drift monotonicity	5	deterministic_entropy_drift, entropy_increases_under_degradation	Proved
Survival product formula	5	survival_product_formula	Proved
Hazard–survival relation	5	hazard_survival_relation	Proved
Survival monotonicity	5	survival_monotone, survival_monotone_decreasing	Proved
Geometric survival	5	geometric_survival	Proved
Infinite product limit	5	bounded_hazard_implies_eventual_stopping, monotone_convergence_pN	Proved
Safe horizon characterization	5	safe_horizon_guarantee_impossible, safe_horizon_is_tacit_dependent	Proved
Tacit irreversibility	5	no_elimination_under_degradation, tacit_evaluation_necessary	Proved
<i>Ch. 6 — Collective and Governance Layer</i>			
Population collapse bound	6	global_degradation_increases_risk	Proved

[**Lean:** Frfp/Core/Kernel.lean], [**Lean:** Frfp/Core/Navigation.lean], [**Lean:** Frfp/Core/Confluence.lean], [**Lean:** Frfp/Core/Grothendieck.lean], [**Lean:** Frfp/Core/Semantics.lean], [**Lean:** Frfp/Core/SemanticCorrectness.lean], [**Lean:** Frfp/Core/DynamicLayer.lean], [**Lean:** Frfp/Core/ProbabilityProven.lean], [**Lean:** Frfp/Core/TacitDependence.lean], [**Lean:** Frfp/Core/InstitutionalLayer.lean], [**Lean:** Frfp/Core/CollectiveLayer.lean], [**Lean:** Frfp/Core/Governance.lean]

Appendix B Appendix B: Module Architecture and Dependency Map

Overview

The FRFP Lean library is structured in five dependency layers. No module imports a module from a higher layer. The dependency graph is acyclic; the build system (Lake) verifies this property.

Layer Dependency Summary

Layer	Modules	Depends on
L1 Kernel	KernelDefinitions	Mathlib only
L2 Float	FloatTheory, FloatMonotonicity	Mathlib, L1
L3 Structure	CategoryStructure, PipelineComposition, GrothConstruction, Termination	L1, L2, Mathlib
L4 Semantics	SemanticCorrectness, TacitDependence, ExplicitArtifact, ReductionSemantics, ConfluenceProof, RunsTrajectories	L1, L2, L3, Mathlib
L5 Governance	CollectiveLayer, GovernanceProofs, AmbiguityProtocol, CredenceFramework, DegradationModel, StoppingTime, SurvivalAnalysis, GovernanceSafeExtension	L1–L4, Mathlib

Detailed Import Graph

Module	Direct Imports
KernelDefinitions	Mathlib.Tactic, Mathlib.Data.Finset.Basic
FloatTheory	KernelDefinitions, Mathlib.Analysis.SpecialFunctions.Pow.Real
FloatMonotonicity	FloatTheory
CategoryStructure	KernelDefinitions, Mathlib.CategoryTheory.Category.Basic
PipelineComposition	CategoryStructure
GrothConstruction	CategoryStructure, Mathlib.CategoryTheory.Grothendieck
Termination	KernelDefinitions, Mathlib.Order.WellFounded
ReductionSemantics	Termination, KernelDefinitions
ConfluenceProof	Termination, ReductionSemantics
SemanticCorrectness	ReductionSemantics, GrothConstruction
TacitDependence	KernelDefinitions
ExplicitArtifact	ReductionSemantics
RunsTrajectories	SemanticCorrectness
CredenceFramework	FloatMonotonicity, KernelDefinitions
DegradationModel	CredenceFramework
StoppingTime	CredenceFramework, Mathlib.Probability.StoppingTime
SurvivalAnalysis	StoppingTime
CollectiveLayer	CredenceFramework, SemanticCorrectness
GovernanceProofs	CollectiveLayer
AmbiguityProtocol	GovernanceProofs
GovernanceSafeExtension	GovernanceProofs

Root and Top-Level Files

File	Role
Frfp.lean	Root re-export: imports all 21 core modules
FRFPReport.lean	Executable theorem inventory report
lakefile.lean	Build configuration
lean-toolchain	Pins Lean version
lake-manifest.json	Locks exact Mathlib commit

Minimal Module Dependency Map

The three minimal modules compile independently of the full core:

Module	Imports
<code>Frfp/Minimal/MinimalCore.lean</code>	<code>Mathlib.Tactic</code> only
<code>Frfp/Minimal/MinimalBasis.lean</code>	<code>MinimalCore</code>
<code>Frfp/Minimal/MinimalReport.lean</code>	<code>MinimalBasis</code>

This independence is intentional. The minimal basis can be used as a lightweight dependency for projects that only need the 7-primitive FRFP foundation.

Theorem Count by Layer

Layer	Theorems	Key topics
L1 Kernel	8	Object types, primitives, kernel invariants
L2 Float	31	IEEE 754 arithmetic, credence bounds
L3 Structure	46	Category axioms, composition, Grothendieck, termination
L4 Semantics	70	Reduction, confluence, denotational semantics, bijection
L5 Governance	114	Probability, degradation, collective layer, governance
Total (core)	269	
Minimal	15	

Appendix C Appendix C: Theory Incorporation Proposals

This appendix provides the full text of all eleven theory incorporation proposals generated during the FRFP formalization process. Each proposal is a recommended change to the theory paper(s) to align the informal presentation with the formal Lean development.

Proposal P-01

Title: Reduction operates on term objects, not category objects.

Problem: The FRFP theory paper describes reduction as operating on “objects in the explicit category E_{cat} .” In the Lean formalization, this caused a type error: categorical objects are not terms, and reduction operates on terms in a term calculus.

Proposed fix: Add a section to the theory paper distinguishing the term calculus (the syntactic language on which reduction is defined) from the categorical semantics (the mathematical universe in which the denotational meanings of terms live). State explicitly that reduction arrows are term-level, while functors are category-level.

Impact: Chapter 3 (reduction), Chapter 4 (semantics).

Proposal P-02

Title: The run-to- ω bijection requires an explicit construction.

Problem: The theory paper describes the run-to- ω correspondence informally as “a natural correspondence.” In Lean, we had to construct explicit functions `runToOmega` and `omegaToRun` and prove two round-trip properties.

Proposed fix: Add a formal definition of the bijection in the theory paper, stating:

$$\text{runTo}\omega : \text{Trace} \rightarrow \Omega \quad \text{and} \quad \omega\text{ToRun} : \Omega \rightarrow \text{Trace}$$

with round-trip equations $\omega\text{ToRun} \circ \text{runTo}\omega = \text{id}$ and $\text{runTo}\omega \circ \omega\text{ToRun} = \text{id}$.

Impact: Chapter 4 (semantics).

Proposal P-03

Title: Credence clamping is a normative design choice, not a consequence.

Problem: The theory paper implies that credence values are “naturally” bounded in $[0, 1]$. In Lean, we discovered that values outside $[0, 1]$ are type-valid without an explicit clamping operation. Clamping is a design choice, not a mathematical consequence.

Proposed fix: Define the clamping function explicitly:

$$\text{clamp}(x) = \max(0, \min(1, x))$$

and state that all credence computations apply `clamp` after each arithmetic step. Justify the choice normatively.

Impact: Chapter 5 (dynamic layer).

Proposal P-04

Title: Navigation equivalence must be explicitly defined as an equivalence relation.

Problem: The theory paper uses $\sim_{\text{nav}}$ without stating reflexivity, symmetry, and transitivity. Lean requires all three proofs explicitly.

Proposed fix: Add a definition block stating:

$$t \sim_{\text{nav}} t \quad (\text{reflexivity}), \quad t \sim_{\text{nav}} t' \Rightarrow t' \sim_{\text{nav}} t \quad (\text{symmetry}), \quad t \sim_{\text{nav}} t'' \text{ if } t \sim_{\text{nav}} t' \sim_{\text{nav}} t''.$$

Impact: Chapter 3 (reduction), Chapter 4 (semantics).

Proposal P-05

Title: Normal forms are unique modulo navigation equivalence, not absolutely unique.

Problem: The theory paper claims “unique normal forms.” In Lean, we proved uniqueness only up to $\sim_{\text{nav}}$. Two terms can have the same observable behavior but different navigation paths; their normal forms are navigation-equivalent but not definitionally equal.

Proposed fix: Correct the uniqueness claim to read “unique normal form up to navigation equivalence,” and state the theorem explicitly.

Impact: Chapter 3 (reduction).

Proposal P-06

Title: Governance safe extension requires a step-compatibility precondition.

Problem: Without a compatibility precondition, a malformed step could be appended to a well-formed trace, invalidating the extension. The step-compatibility predicate must be defined and included in the theorem statement.

Proposed fix: Define `StepCompatible(trace, step)` as the predicate that the new step’s intent and authorization are coherent with the preceding context. State the safe extension theorem explicitly.

Impact: Chapter 6 (governance), new theorem section.

Proposal P-07

Title: First-witness canonicity should be stated as a numbered theorem.

Problem: The theory paper notes informally that governance failures have a “first occurrence.” This deserves formal theorem status since it is the foundation for constructive governance monitoring algorithms.

Proposed fix: Add a theorem: “For any malformed governance trace, there exists a unique minimal index k^* such that step k^* is the first governance violation.”

Impact: Chapter 6 (governance), new results section.

Proposal P-08

Title: Consensus impossibility requires a typed no-grounding precondition.

Problem: The original consensus theorem admitted unintended models where agents could achieve consensus by vacuous means. A typed `groundingRequired : Bool` precondition prevents this collapse.

Proposed fix: Add the grounding-required type flag as an explicit hypothesis in the consensus impossibility statement.

Impact: Chapter 6 (collective layer).

Proposal P-09

Title: The Grothendieck construction requires a split-opfibration condition.

Problem: The theory paper applies the Grothendieck construction without mentioning the split-opfibration requirement on the indexing functor. Lean required this structure explicitly.

Proposed fix: Add a brief categorical prerequisite section noting that the FRFP semantic functor $F : \mathbf{C}_0 \rightarrow \mathbf{Cat}$ must be a split opfibration. Cite Mac Lane (1971) Ch. XII.

Impact: Chapter 4 (semantics).

Proposal P-10

Title: Phase partition well-definedness requires finiteness of pipeline terms.

Problem: The phase partition theorem holds only for finite pipeline terms. Lean's type-checker requires a finiteness hypothesis; without it, the proof does not go through.

Proposed fix: Add "for any finite pipeline term t " as a hypothesis to the phase partition theorem. Note that infinite pipelines are not currently modeled.

Impact: Chapter 1 (kernel), Chapter 3 (reduction).

Proposal P-11

Title: Stopping time measurability requires explicit filtration structure.

Problem: The theory paper cites stopping times without specifying the filtration $(\mathcal{F}_t)_{t \geq 0}$ with respect to which measurability is claimed. Lean's Mathlib requires the filtration as an explicit argument.

Proposed fix: Add a definition of the natural FRFP filtration: $\mathcal{F}_t = \sigma(\text{all protocol events up to time } t)$, and state measurability with respect to this filtration.

Impact: Chapter 5 (dynamic layer).

Appendix D Appendix D: Axiom Citation Guide

This appendix annotates the primary sources cited by axioms in the Lean build. Every axiom in the active build cites one of these sources; the right-hand column notes which Lean modules carry those citations.

Primary Sources Cited by Axioms

The following references are cited within the Lean source code as axiom justifications. Every axiom in the active build cites one of these sources.

`oindentIEEE Standard 754-2019` [9]. *IEEE Standard for Floating-Point Arithmetic*. IEEE Computer Society, 2019. *Covers*: §4 (attributes), §5 (operations), §6 (infinity/NaN), §10 (total ordering). *Cited by*: 59 axioms in `FloatTheory` and `FloatMonotonicity`.

`oindentBillingsley, P.` [4] (1995). *Probability and Measure* (3rd ed.). John Wiley & Sons. *Covers*: probability space axioms (Ch. 1), conditional probability (Ch. 5), stopping times (Ch. 7), measure theory (Ch. 3). *Cited by*: 21 axioms in `CollectiveLayer`, `StoppingTime`, `SurvivalAnalysis`.

`oindentBaader, F., & Nipkow, T.` [2] (1998). *Term Rewriting and All That*. Cambridge University Press. *Covers*: abstract reduction systems (Ch. 1–2), termination (Ch. 2–3), confluence/Church-Rosser (Ch. 2–3), normal forms (Ch. 2). *Cited by*: 18 axioms in `Termination`, `ReductionSemantics`, `ConfluenceProof`.

`oindentMac Lane, S.` [1] (1971). *Categories for the Working Mathematician*. Springer. *Covers*: category axioms (Ch. I), functors (Ch. II), natural transformations (Ch. III), Grothendieck construction (Ch. XII). *Cited by*: 7 axioms in `CategoryStructure`, `GrothConstruction`.

`oindentDurrett, R.` [5] (2019). *Probability: Theory and Examples* (5th ed.). Cambridge University Press. *Covers*: martingales (Ch. 4), stopping times (Ch. 4), convergence (Ch. 5). *Cited by*: 4 axioms in `StoppingTime`, `SurvivalAnalysis`.

`oindentNewman, M.H.A.` [3] (1942). On theories with a combinatorial definition of “equivalence.” *Annals of Mathematics*, 43(2), 223–243. *Covers*: Church-Rosser property, confluence from termination and local confluence. *Cited by*: 2 axioms in `ConfluenceProof`.

`oindentRudin, W.` [6] (1976). *Principles of Mathematical Analysis* (3rd ed.). McGraw-Hill. *Covers*: metric spaces (Ch. 2), bijection properties (Ch. 2). *Cited by*: 2 axioms in `SemanticCorrectness`.

`oindentDiestel, R.` [8] (2010). *Graph Theory* (4th ed.). Springer. *Covers*: directed graphs (Ch. 1), paths and reachability (Ch. 1). *Cited by*: 1 axiom in a helper module.

`oindentCover, T.M., & Thomas, J.A.` [7] (2006). *Elements of Information Theory* (2nd ed.). John Wiley & Sons. *Covers*: Shannon entropy (Ch. 2), entropy of discrete distributions. *Cited by*: 1 axiom in `CollectiveLayer` (entropy drift).

Framework Sources (Theory Premises)

FRFP Theory Paper. *FRFP: A Formal Theory of Explicit–Tacit Computation*. Author. 2024/2025. *Source of*: all 7 FRFP base axioms and their module-level copies (total 40 axioms).

`oindentPolanyi, M.` [10] (1966). *The Tacit Dimension*. Routledge. *Referenced in justification for*: HEG and HEC axioms (tacit knowledge is inaccessible to explicit reconstruction).

Tool References

Lean 4 [11]. <https://leanprover.github.io/>

Mathlib4 [12]. https://leanprover-community.github.io/mathlib4_docs/

Appendix E Appendix E: Naming Conventions

This appendix documents the naming conventions used throughout the FRFP Lean formalization. Consistent naming enables grep-based navigation and prevents name collisions across 21 modules.

General Principles

1. **Lean 4 style:** Follow Lean 4 community naming (snake_case for definitions, theorems, and axioms; CamelCase for types and structures).
2. **Layer prefix:** Module-specific constants are prefixed with a short module abbreviation (e.g., float_, sem_, gov_).
3. **Source annotation:** Every axiom has a comment on the preceding line: - Reference: [Author Year], §X.Y.
4. **Novel marker:** Novel theorems (those not present in published papers) carry a comment - Novel: not in source papers.

Type and Structure Naming

Pattern	Example	Rule
Kernel object types	Object, Primitive	CamelCase, bare name
Categorical objects	ExplicitObject, TacitObject	CamelCase, full adjective
Category types	Ecat, Tcat	Short, subscript rendered as plain text
Structures	ProtocolStep, GovernanceTrace	CamelCase compound
Predicates (Bool-valued)	AIExecutable, GovWF	CamelCase; WF = well-formed
Propositions (Prop-valued)	isGrounded, isNormal	camelCase with is- prefix

Definition and Function Naming

Pattern	Example	Rule
Semantic function	sem	Lowercase, bare
Observation function	observe	Lowercase, bare
Bijection functions	runToOmega, omegaToRun	camelCase with direction
Clamping function	credenceClamp	camelCase, noun-verb
Phase function	phaseOf	camelCase, noun-Of
Pipeline constructor	mkPipeline	camelCase, mk- prefix

Axiom Naming

Pattern	Example	Rule
FRFP premise axioms	heg_axiom, il_axiom	lowercase abbrev + _axiom
IEEE 754 axioms	float_add_comm	float_ prefix + property
Probability axioms	survival_product_formula	noun + _formula / _property
Rewriting axioms	reduction_wf, newman_lemma	lowercase, domain-specific
Category axioms	ecat_assoc, tcat_id_left	category abbreviation + property

Theorem and Lemma Naming

Pattern	Example	Rule
Uniqueness theorems	ets_unique, rb_uniqueness	subject + _unique / _uniqueness
Existence theorems	normal_form_exists	noun + _exists
Invariance theorems	observation_invariance	noun + _invariance
Monotonicity theorems	degradation_monotone	noun + _monotone
Impossibility theorems	tacit_not_reconstructable	subject + not + predicate
Novel theorems	first_unresolved_witness	descriptive, no abbreviation
Bijection theorems	runToOmega_left_inverse	function name + _left/right_inverse

Module and File Naming

Module	File	Convention
Core modules	<code>Frfp/Core/CamelCaseName.lean</code>	CamelCase, noun phrase
Minimal modules	<code>Frfp/Minimal/MinimalXxx.lean</code>	Minimal + topic
Example modules	<code>Example/CamelCase.lean</code>	CamelCase, real-world name
Root file	<code>Frfp.lean</code>	Framework name only
Report file	<code>FRFPReport.lean</code>	ALL-CAPS acronym + Report

Symbol Conventions in the Mathematical Notation

Symbol	Meaning	Lean identifier
$\mathbf{C}_0$	The 3-object kernel category	<code>KernelCategory</code>
$\mathbf{E}_{\text{cat}}$	Explicit object category	<code>Ecat</code>
$\mathbf{T}_{\text{cat}}$	Tacit object category	<code>Tcat</code>
E_0	Explicit object	<code>Object.Explicit</code>
T_0	Tacit object	<code>Object.Tacit</code>
$\int F$	Grothendieck construction	<code>GrothConstruction</code> <code>frfpFunctor</code>
$\text{runTo}\omega$	Run-to-omega map	<code>runToOmega</code>
ωToRun	Omega-to-run map	<code>omegaToRun</code>
$\sim_{\text{nav}}$	Navigation equivalence	<code>NavEquiv</code>
Sem	Semantic function	<code>sem</code>
obs	Observation function	<code>observe</code>
GovWF	Governance well-formedness	<code>GovWF</code>
k^*	First unresolved witness	<code>k_star</code> (in proof context)

Comment Header Format

Every Lean file begins with a header block:

```

/-!
# Module Name

Brief description of the module's role in the FRFP formalization.

## Dependencies
- List of imported modules

## Key results
- Theorem 1: brief statement
- Theorem 2: brief statement

## Axiom sources
- Reference 1
- Reference 2
-/

```

Deprecated Naming Patterns

The following patterns were used in early development and should not appear in the final build:

Deprecated pattern	Replacement
<code>frfp_XXX</code> prefix on everything	Layer-specific prefix only
<code>h1</code> , <code>h2</code> , <code>h3</code> as hypothesis names	Descriptive names (<code>h_wf</code> , <code>h_gt</code>)
<code>Axiom1</code> , <code>Axiom2</code> enumeration	Full descriptive name
<code>sorry</code> as placeholder beyond draft	Tracked in issue list; not committed

Appendix F Appendix F: Theory-to-Lean Correspondence

This appendix provides a complete traceability record mapping every named theorem and lemma in the FRFP Theory Paper to its machine-checked counterpart in the Lean formalization. It also lists Lean-only results that have no direct theory-paper antecedent.

Legend

Status	Meaning
Proved	Fully proved in Lean; no <code>sorry</code>
Lean-only	Proved in Lean; not stated in Theory Paper
Novel	New result discovered through formalization
Demoted	Was an axiom in earlier draft; now a proved theorem

Chapter 1 — The Kernel Category

Theory Paper Claim	§	Lean Name	Status
Explicit–Tacit Separation	1.4	<code>no_morphism_tacit_to_explicit</code>	Proved
		<code>no_morphism_T0_to_E0</code>	Proved
Uniqueness of the boundary morphism	1.4	<code>RB_unique_boundary</code>	Proved
AI executability partition	1.4	<code>AI_explicit_only</code>	Proved
		<code>AI_preserves_explicit</code>	Proved
		<code>human_only_not_AI</code>	Proved
Composition preserves typing	1.5	<code>explicit_composition_closed</code>	Proved
Phase partition	1.6	<code>primitive_completeness</code>	Proved
		<code>primitive_well_typed</code>	Proved
Boundary uniqueness by phase	1.6	<code>RB_is_boundary</code>	Proved
No tacit-to-explicit return	1.7	<code>no_morphism_tacit_to_explicit</code>	Proved
		<code>RI_is_explicit, TE_is_tacit</code>	Proved

[Lean: Frfp/Core/Kernel.lean]

Chapter 2 — The Reduced Axiom Basis

Theory Paper Claim	§	Lean Name	Status
ETS is a theorem (demoted)	2.2	<code>no_morphism_tacit_to_explicit</code>	Demoted
RB uniqueness is a theorem (demoted)	2.2	<code>RB_unique_boundary</code>	Demoted
AE is a theorem (demoted)	2.2	<code>AI_explicit_only</code>	Demoted
MD is a theorem (demoted)	2.2	<code>degradation_monotone</code>	Demoted

[Lean: Frfp/Core/Kernel.lean], [Lean: Frfp/Core/DynamicLayer.lean]

Chapter 3 — Reduction and Confluence

Theory Paper Claim	§	Lean Name	Status
Termination	3.5	<code>combined_reduction_properties</code>	Proved
Local confluence of $\rightarrow_c$	3.6	<code>combined_reduction_properties</code>	Proved
Confluence (Church–Rosser)	3.7	<code>combined_reduction_properties</code>	Proved
		<code>nav_confluent</code>	Proved
Normal form uniqueness mod $\equiv_{\nabla}$	3.8	<code>unique_normal_forms</code>	Proved
		<code>normal_form_unique</code>	Proved
		<code>nav_equiv_equivalence</code>	Proved
Observable invariance	3.9	<code>semantic_correctness_theorem</code>	Proved
		<code>obsCorrect_invariant</code>	Proved
Protocol normal form	3.10	<code>normal_forms_exist_unique</code>	Proved

[**Lean:** Frfp/Core/Confluence.lean], [**Lean:** Frfp/Core/Navigation.lean], [**Lean:** Frfp/Core/SemanticCorrectness.lean]

Chapter 4 — Denotational and Operational Semantics

Theory Paper Claim	§	Lean Name	Status
Reduction induces morphism	4.3	reduction_induces_morphism	Proved
		reduction_on_objects	Proved
Denotational–operational agreement	4.8	semantic_functor_properties	Proved
		sem_respects_reduction	Proved
Navigation equivalence is an equiv. relation	4.9	nav_equiv_equivalence	Proved
		nav_equiv_refl, nav_equiv_symm, nav_equiv_trans	Proved
Pipeline immutability	4.10	reduction_system_well_defined	Proved

[**Lean:** Frfp/Core/Grothendieck.lean], [**Lean:** Frfp/Core/Semantics.lean], [**Lean:** Frfp/Core/SemanticCorrectness.lean], [**Lean:** Frfp/Core/OperationalSemantics.lean]

Chapter 5 — Dynamic and Probabilistic Layer

Theory Paper Claim	§	Lean Name	Status
Degradation monotonicity	5.3	degradation_monotone	Proved
		degradation_contractive	Proved
Entropy drift monotonicity	5.4	deterministic_entropy_drift	Proved
		entropy_increases_under_degradation	Proved
Survival product formula	5.5	survival_product_formula	Proved
Hazard–survival relation	5.5	hazard_survival_relation	Proved
Survival monotonicity	5.5	survival_monotone	Proved
		survival_monotone_decreasing	Proved
Geometric survival	5.5	geometric_survival	Proved
Infinite product limit	5.5	bounded_hazard_implies_eventual_stopping	Proved
		monotone_convergence_pN	Proved
Safe horizon characterization	5.6	safe_horizon_guarantee_impossible	Proved
		safe_horizon_is_tacit_dependent	Proved
		safeHorizon_monotone_in_states	Proved
Tacit irreversibility	5.7	no_elimination_under_degradation	Proved
		tacit_evaluation_necessary	Proved

[**Lean:** Frfp/Core/DynamicLayer.lean], [**Lean:** Frfp/Core/ProbabilityProven.lean], [**Lean:** Frfp/Core/TacitDependence.lean]

Chapter 6 — Collective and Governance Layer

Theory Paper Claim	§	Lean Name	Status
Population collapse bound	6.4	global_degradation_increases_risk	Proved
		no_fully_automated_governance	Proved
No grounding by consensus	6.5	consensus_no_grounding_creation	Proved
		consensus_requires_tacit_oracle	Proved
		no_automated_consensus_guarantee	Proved
Governance safe extension	6.6	governance_non_automation	Proved
		oversight_necessity_theorem	Proved
Collective entropy drift	6.7	degradation_aggregation_commute	Proved

[**Lean:** Frfp/Core/InstitutionalLayer.lean], [**Lean:** Frfp/Core/CollectiveLayer.lean], [**Lean:** Frfp/Core/TacitDependence.lean], [**Lean:** Frfp/Core/Governance.lean]

Chapter 7 — Novel Results from Formalization

Theory Paper Claim	§	Lean Name	Status
First unresolved witness	7.1	hallucination_is_tacit_dependent	Novel
		hallucination_detection_impossible	Novel
Governance safe extension (novel)	7.2	governance_non_automation	Novel
		oversight_necessity_theorem	Novel
Protocol normal form (novel)	7.3	normal_forms_exist_unique	Novel
Tacit source classification	7.4	TE_is_tacit, HFD_is_tacit	Proved
Tacit source stays tacit	7.4	construct_tacit_population_is_tacit	Proved
AI steps never target tacit space	7.4	AI_explicit_only	Proved
AI/human disjoint	7.4	human_only_not_AI	Proved
Unique boundary entry	7.4	RB_unique_boundary	Proved
IL pairwise intent consistency	7.4	institutional_layer_requirements_satisfied	Proved
AR: no unresolved ambiguity	7.4	dynamic_layer_requirements_satisfied	Proved
CSC: lock is monotone	7.4	tacitState_le_refl, tacitState_le_trans	Proved

[**Lean:** Frfp/Core/Kernel.lean], [**Lean:** Frfp/Core/TacitDependence.lean], [**Lean:** Frfp/Core/Navigation.lean], [**Lean:** Frfp/Core/Governance.lean]

Lean-Only Results (No Theory Paper Antecedent)

These theorems were proved in Lean as part of establishing the formalization but do not correspond to a named claim in the Theory Paper. They are either auxiliary lemmas, module-requirement witnesses, or results that emerged from proof obligations and have not yet been incorporated into the Theory Paper.

Lean Name	Module	Description
runTo ω _preserves_admissibility	OperationalSemantics	runToOmega bijection preserves admissibility
runs_are_measurable	OperationalSemantics	Stochastic runs are measurable
stochastic_quantities_well_defined	OperationalSemantics	Well-definedness of the probability space
RB_sharp_consistent_with_RB	Semantics	Sharp representational backflow agrees with RB
RB_sharp_explicit_to_tacit	Semantics	Sharp RB maps explicit to tacit
sem_nav_mono	Semantics	Semantic functor is monotone under navigation
obsCorrect_preserved_by_nav	Semantics	Observational correctness stable under navigation
obsCorrect_preserved_by_reduction	Semantics	Observational correctness stable under reduction
different_obsCorrect_different_nf_classes	SemanticCorrectness	Distinct correctness classes have distinct NF classes
obsCorrect_class_invariant	SemanticCorrectness	Correctness class is an invariant
sameNF_refl/symm/trans	ExplicitArtifact	NF equivalence is a congruence
sameSyn_implies_sameNF	ExplicitArtifact	Syntactic equality implies NF equality
syn_refines_nf	ExplicitArtifact	Syntactic reduction refines NF
degradation_aggregation_commute	InstitutionalLayer	Degradation commutes with aggregation
stability_not_correctness	InstitutionalLayer	Stability does not imply correctness
consensus_instability_without_grounding	InstitutionalLayer	Consensus without grounding is unstable
confidence_inflation_preserves_grounding	DynamicLayer	Inflation does not create tacit grounding
accelerated_inevitability_with_feedback	DynamicLayer	Feedback accelerates hallucination inevitability
redundancy_extends_horizon	CollectiveLayer	Redundancy extends (but does not eliminate) safe horizon
resilience_connectivity	CollectiveLayer	Population resilience depends on connectivity
grothendieck_category_exists	Grothendieck	The semidirect product $N \ltimes E$ forms a category
grothendieck_assoc	Grothendieck	Associativity of Grothendieck composition
reduction_compose	Grothendieck	Reduction is compositional
navigation_design_decision	Navigation	Documents key design choice for nav. orientation
free_explicit_algebra	TDG	Task Decomposition Grammar forms a free explicit algebra
TE_idempotent, HFD_idempotent	TDG	Task expansion / human-feedback ops are idempotent
TE_HFD_commute	TDG	Task expansion and HFD commute
phase1_initiality	Phase1	Phase-1 kernel is initial among admissible systems
phase1_uniqueness	Phase1	Uniqueness of the Phase-1 functor

Coverage Summary

Category	Theory Claims	Lean Theorems
Ch. 1: Kernel Category	7	14
Ch. 2: Reduced Basis	4	4 (demoted)
Ch. 3: Reduction & Confluence	7	11
Ch. 4: Semantics	4	12
Ch. 5: Dynamic Layer	8	16
Ch. 6: Collective Layer	4	8
Ch. 7: Novel Results	11	14
Total theory-mapped	45	79
Lean-only (no antecedent)	—	27
Grand total	45	106

The 61 additional Lean theorems beyond the 45 theory claims arise from (a) multiple Lean lemmas required to establish a single theory claim, (b) auxiliary well-definedness and requirements-satisfaction witnesses, and (c) the 27 Lean-only results listed above.

References

1. S. Mac Lane. *Categories for the Working Mathematician*. Graduate Texts in Mathematics 5. Springer, 1971.
2. F. Baader and T. Nipkow. *Term Rewriting and All That*. Cambridge University Press, 1998.
3. M. H. A. Newman. On theories with a combinatorial definition of “equivalence.” *Annals of Mathematics*, 43(2):223–243, 1942.
4. P. Billingsley. *Probability and Measure*, 3rd edition. John Wiley & Sons, 1995.
5. R. Durrett. *Probability: Theory and Examples*, 5th edition. Cambridge University Press, 2019.
6. W. Rudin. *Principles of Mathematical Analysis*, 3rd edition. McGraw-Hill, 1976.
7. T. M. Cover and J. A. Thomas. *Elements of Information Theory*, 2nd edition. John Wiley & Sons, 2006.
8. R. Diestel. *Graph Theory*, 4th edition. Graduate Texts in Mathematics 173. Springer, 2010.
9. IEEE Computer Society. *IEEE Standard for Floating-Point Arithmetic*. IEEE Std 754-2019. IEEE, 2019.
10. M. Polanyi. *The Tacit Dimension*. Routledge & Kegan Paul, London, 1966.
11. L. de Moura and S. Ullrich. The Lean 4 theorem prover and programming language. In A. Platzer and G. Sutcliffe, editors, *Automated Deduction – CADE-28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635. Springer, 2021.
12. The Mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020.
13. K. Buzzard, C. Commelin, and P. Massot. Formalising perfectoid spaces. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 299–312. ACM, 2020.
14. T. Dokchitser, V. Dokchitser, C. Maistret, and A. Morgan. Fermat’s last theorem for regular primes. In *14th International Conference on Interactive Theorem Proving (ITP 2023)*, volume 268 of *LIPIcs*, pages 12:1–12:8. Schloss Dagstuhl, 2023.
15. F. van Doorn, S. Kudlyk, J. Seidler, and K. Sugate. A complete formalization of Fermat’s Last Theorem for regular primes in Lean 4. In *Proceedings of the 2025 ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2025)*. ACM, 2025.